Business Process Modeling Notation (**BPMN**)

Pertemuan 7

Business Process

"Business process refer to the set of logically related tasks and behaviors that organizations develop over time to produce specific business results and the unique manner in which these activities are organized and coordinated." (Loudon)

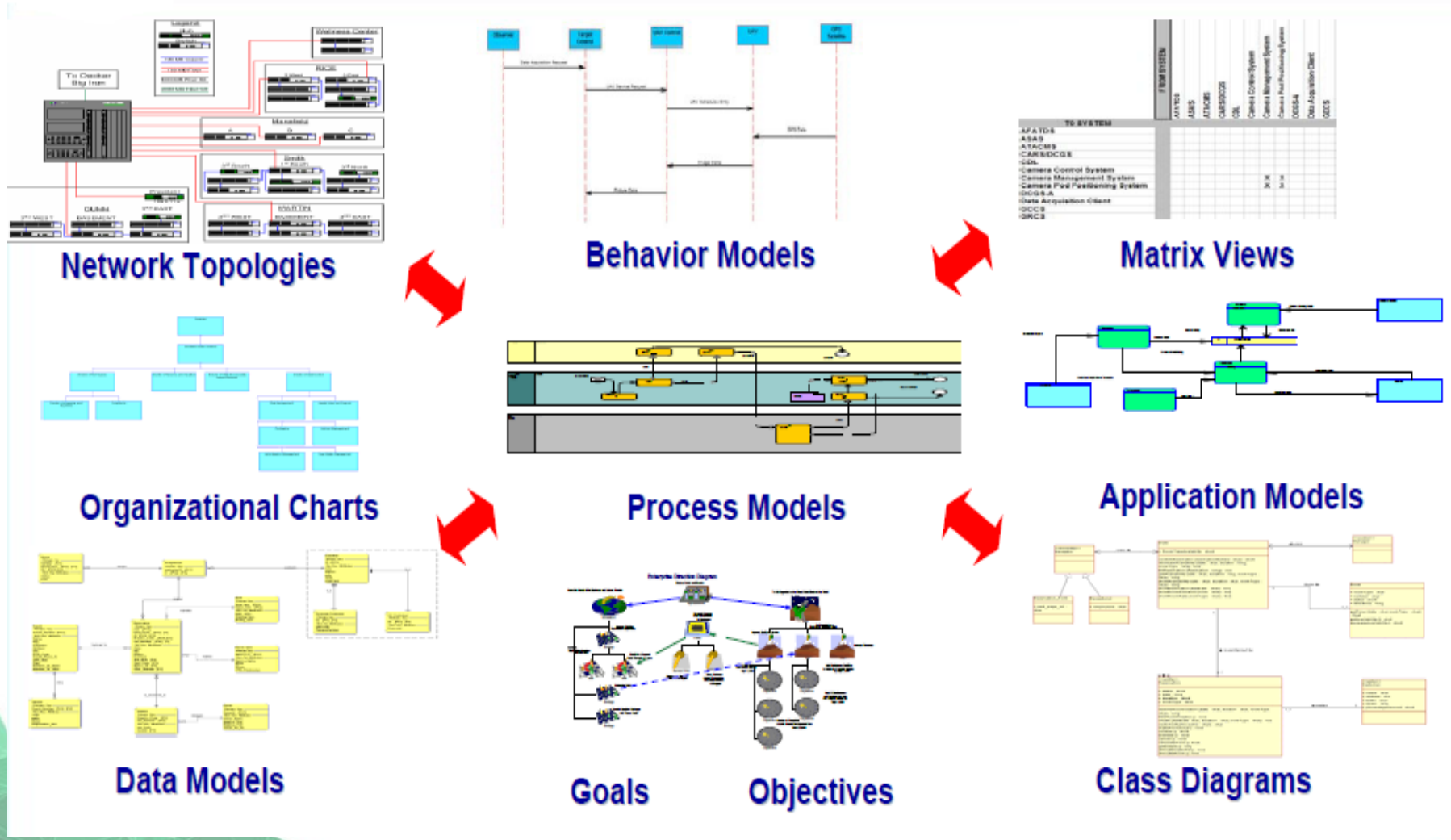
Proses bisnis bisa dikatakan sebagai sekelompok tugas dan aktivitas yang saling berhubungan secara logis, dan diatur serta dikoordinasikan supaya bisa membuat hasil bisnis yang spesifik dan unik.

Business Process

- Semua proses bisnis pada intinya harus bisa mendukung secara optimal proses pengelolaan sumber daya *enterprise*.

- Setiap proses bisnis harus bisa menghasilkan nilai tambah terhadap hasil dari proses bisnis yang dilakukan sebelumnya.

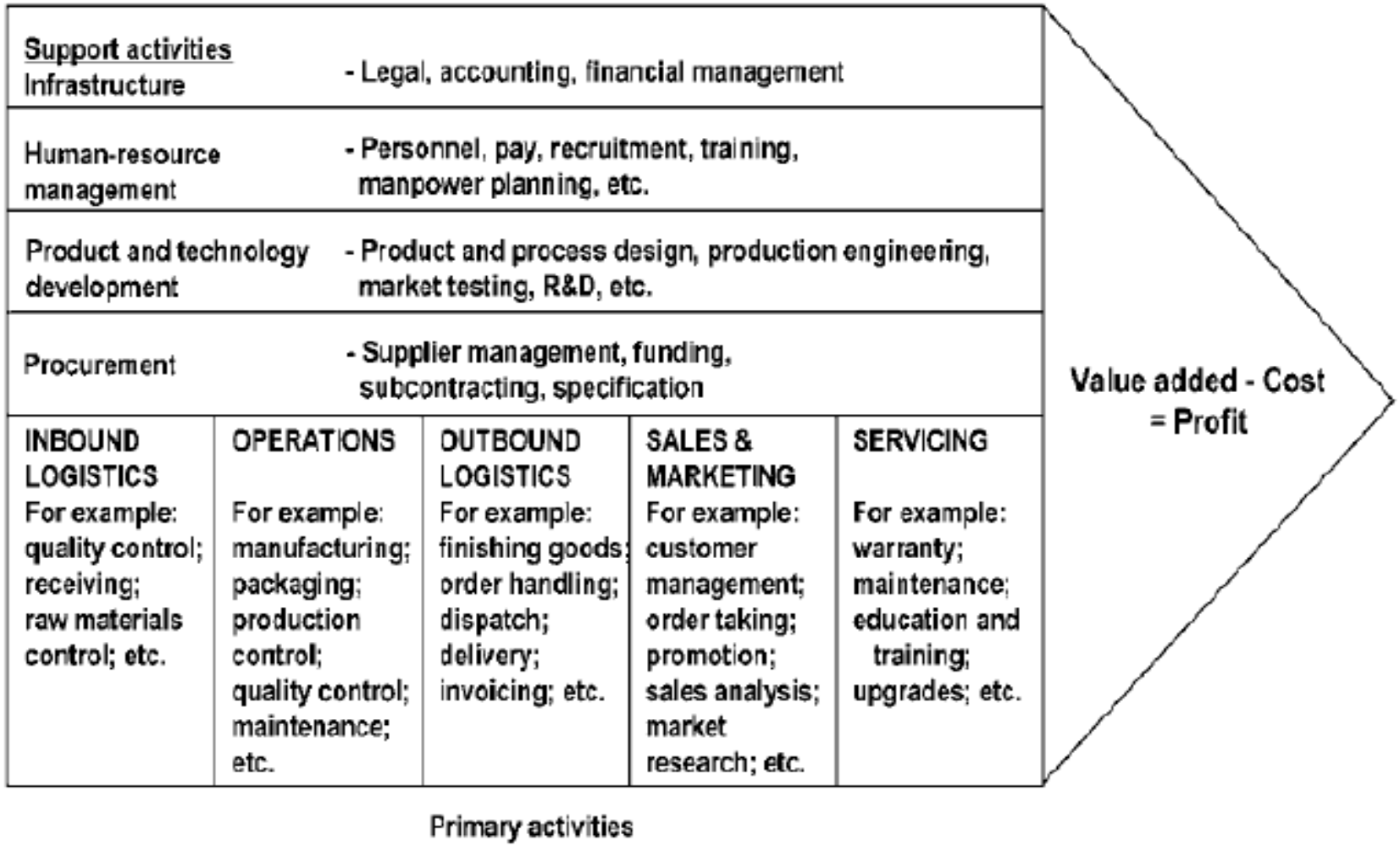
Product of Enterprise Architecture



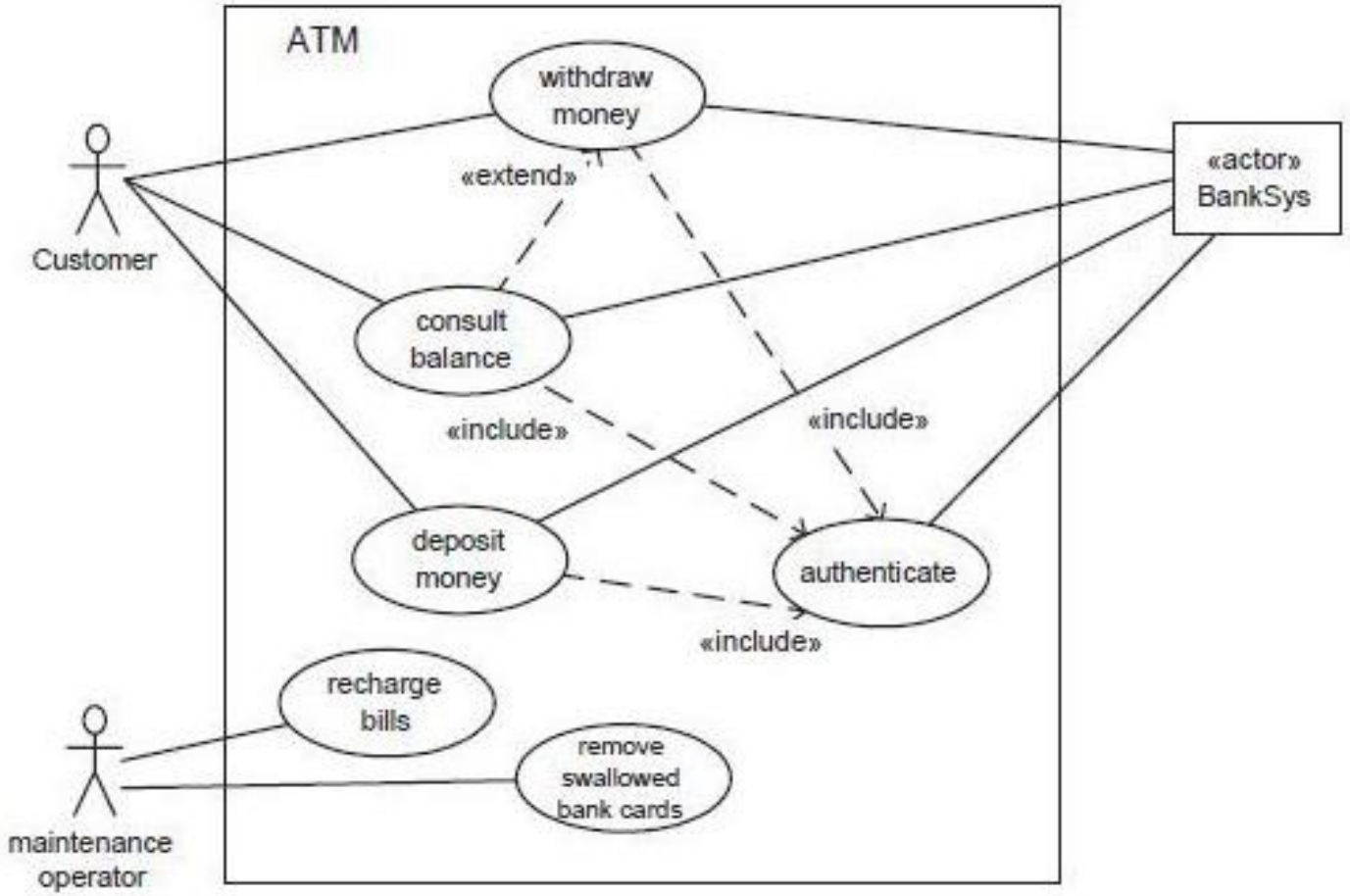
Analisis Business Process

- Value Chain
- Four Stage Life Cycle
  - Bagan Hirarki
- Business Process Modeling Notation (BPMN)
- Use Case Diagram

Value Chain


Support activities
Infrastructure
- Legal, accounting, financial management
Human-resource management
- Personnel, pay, recruitment, training, manpower planning, etc.
Product and technology development
- Product and process design, production engineering, market testing, R&D, etc.
Procurement
- Supplier management, funding, subcontracting, specification
INBOUND LOGISTICS
For example: quality control; receiving; raw materials control; etc.
OPERATIONS
For example: manufacturing; packaging; production control; quality control; maintenance; etc.
OUTBOUND LOGISTICS
For example: finishing goods; order handling; dispatch; delivery; invoicing; etc.
SALES & MARKETING
For example: customer management; order taking; promotion; sales analysis; market research; etc.
SERVICING
For example: warranty; maintenance; education and training; upgrades; etc.
Value added - Cost = Profit
Primary activities

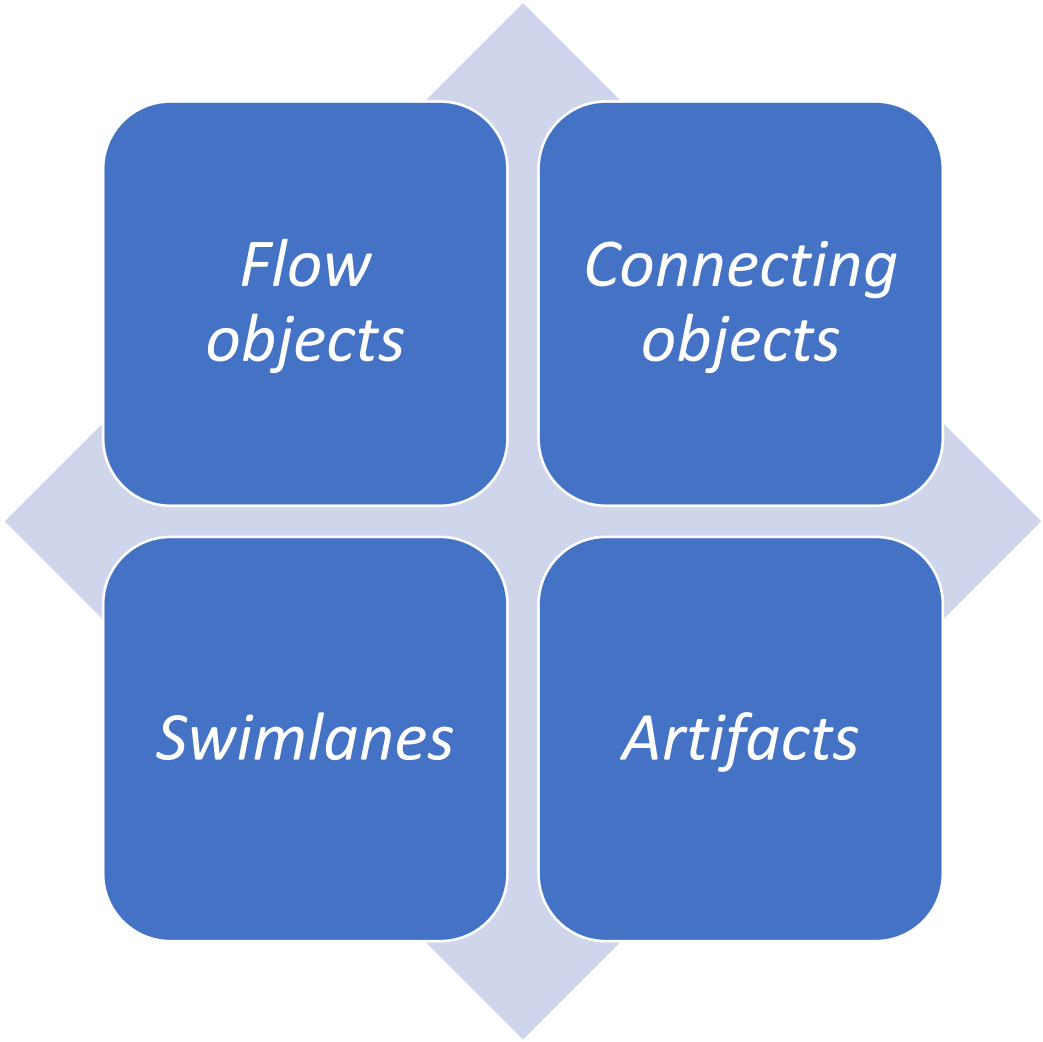
Use Case Diagram



BPMN

- Tahun 2004, BPMN diperkenalkan.
- BPMN» pengembangan dari *flowchart*.
- BPMN merupakan alat bantu (*tool*) berupa notasi gambar untuk menggambarkan atau memvisualisasikan bagaimana proses bisnis yang sedang berlangsung dalam bentuk diagram proses bisnis.
- Tujuan BPMN yaitu menyediakan notasi yang mudah digunakan dan bisa dimengerti oleh semua orang yang terlibat dalam bisnis

4 Elements of BPMN



Flow Objects

Event	Menunjukkan kejadian dari suatu kegiatan dimulai dari *Start*, *Intermediate*, *End*.	
Activity	Menunjukkan aktivitas yang dilakukan oleh suatu proses bisnis.	
Gateway	Fungsinya untuk menunjukkan alur percabangan atau pilihan seperti *Forking*, *Merging*, *Joining*.	

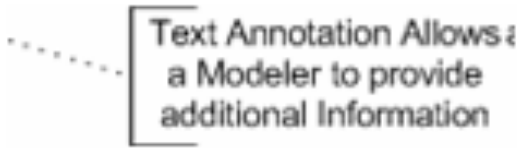
Conntecting Objects

Sequence flow	Menunjukkan urutan aktivitas yang dilakukan dalam suatu proses	
Message flow	Menunjukkan alur pesan yang terjadi antara pelaku proses	
Association	Menunjukkan input atau output yang diperlukan oleh suatu aktivitas	

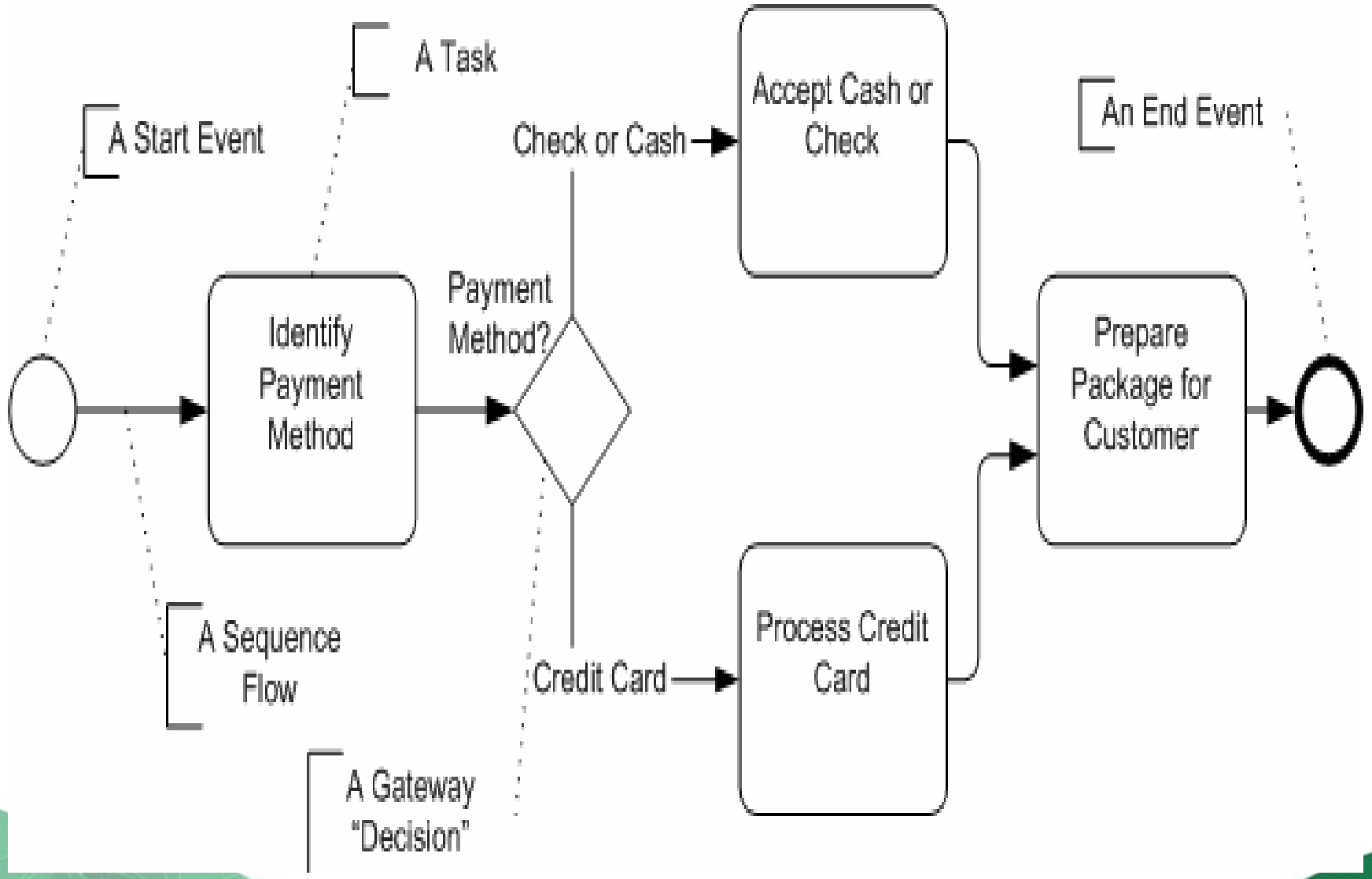
Swimlanes

Pool	Merepresentasikan pelaku suatu aktivitas.	
Lane	Pengelompokan pelaku yang lebih detail.	

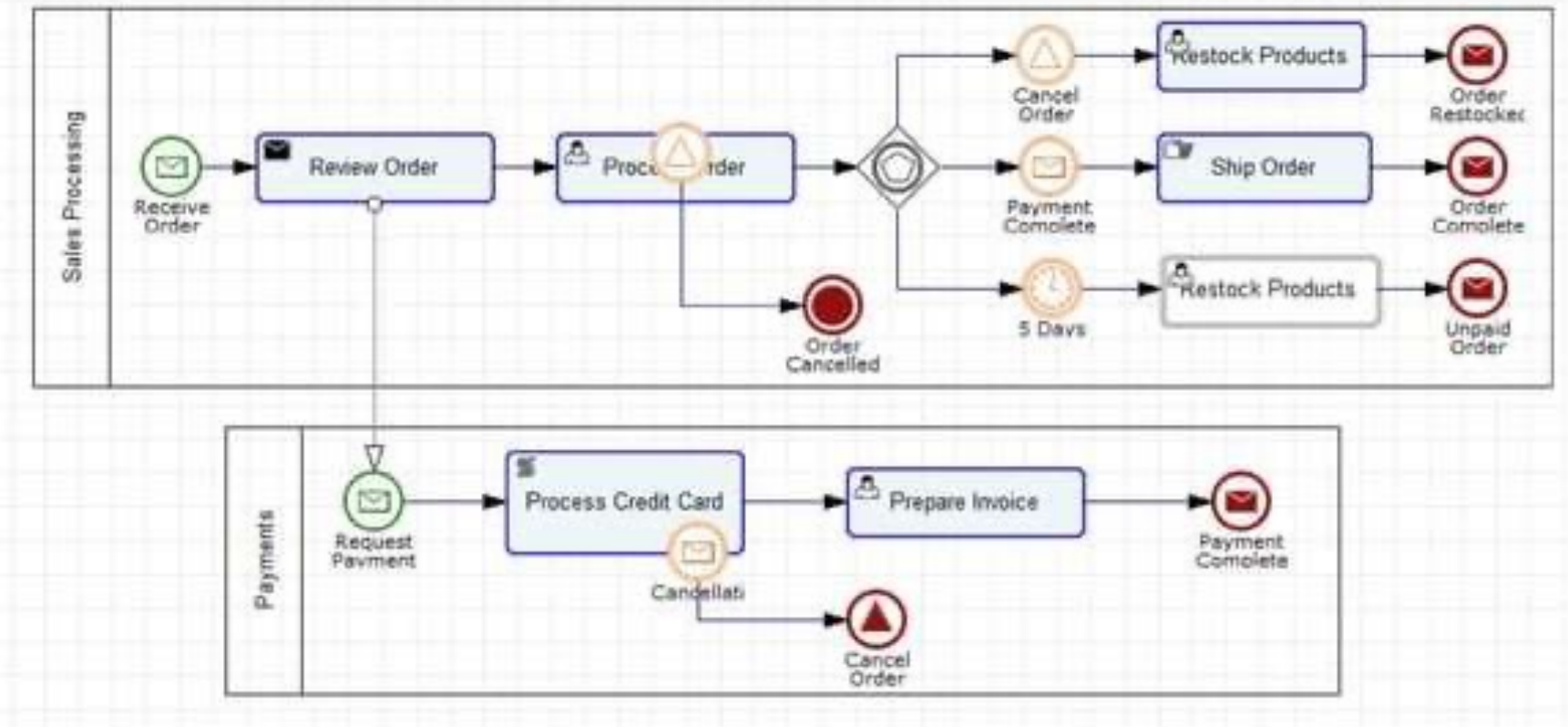
Artifacts

Data object	Menunjukkan data apa yang dibutuhkan oleh suatu aktivitas, atau data apa yang dihasilkan oleh suatu aktivitas.	
Group	Tujuannya untuk lebih memudahkan dalam menganalisis proses dan pendokumentasian. Tidak berpengaruh apa-apa terhadap aktivitas sama sekali.	
Annotation	Merupakan komentar atau deskripsi singkat untuk memperjelas diagram	

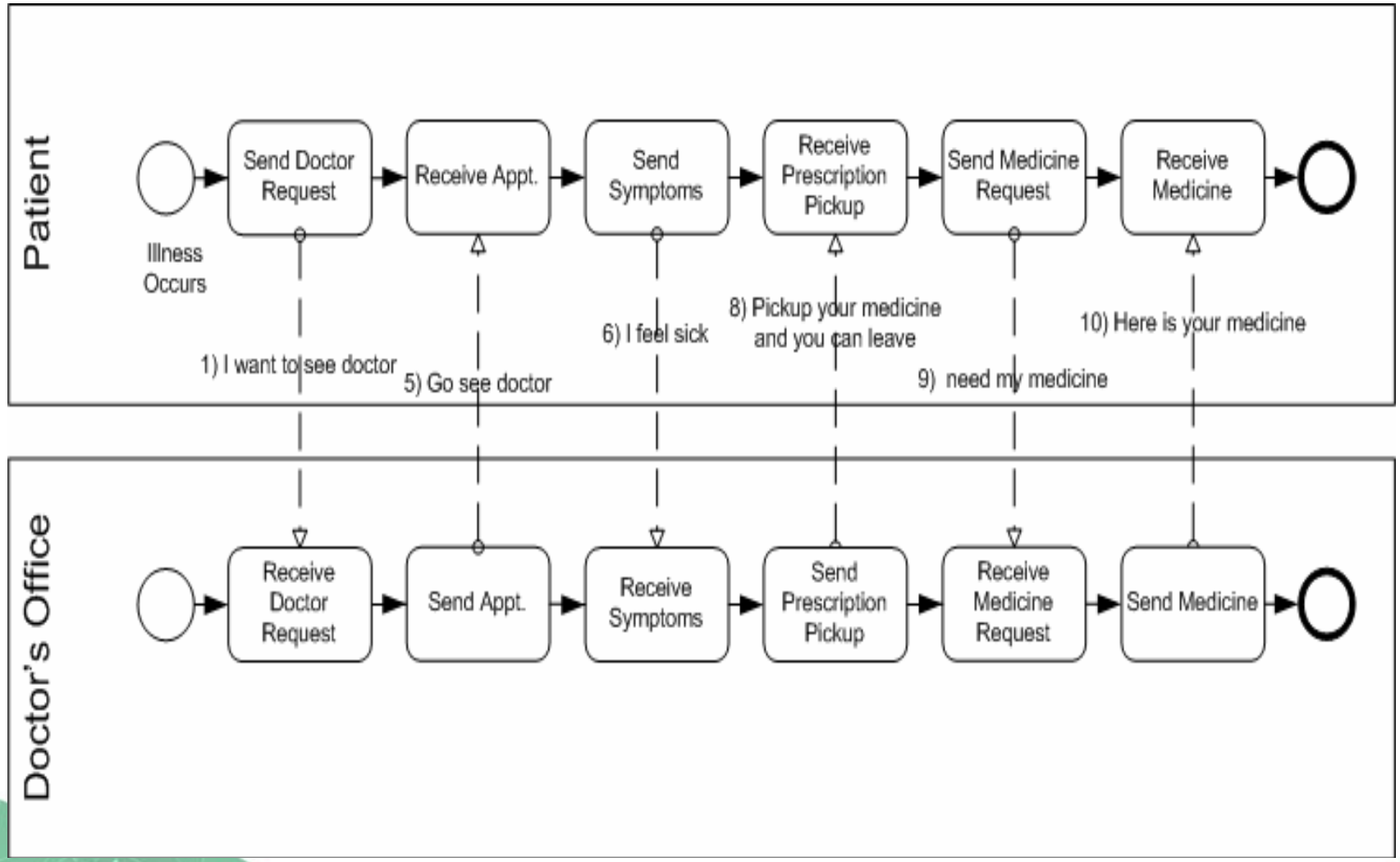
Contoh 1


A Task
A Start Event
Check or Cash
Accept Cash or Check
An End Event
Identify Payment Method
Payment Method?
Prepare Package for Customer
A Sequence Flow
Credit Card
Process Credit Card
A Gateway "Decision"

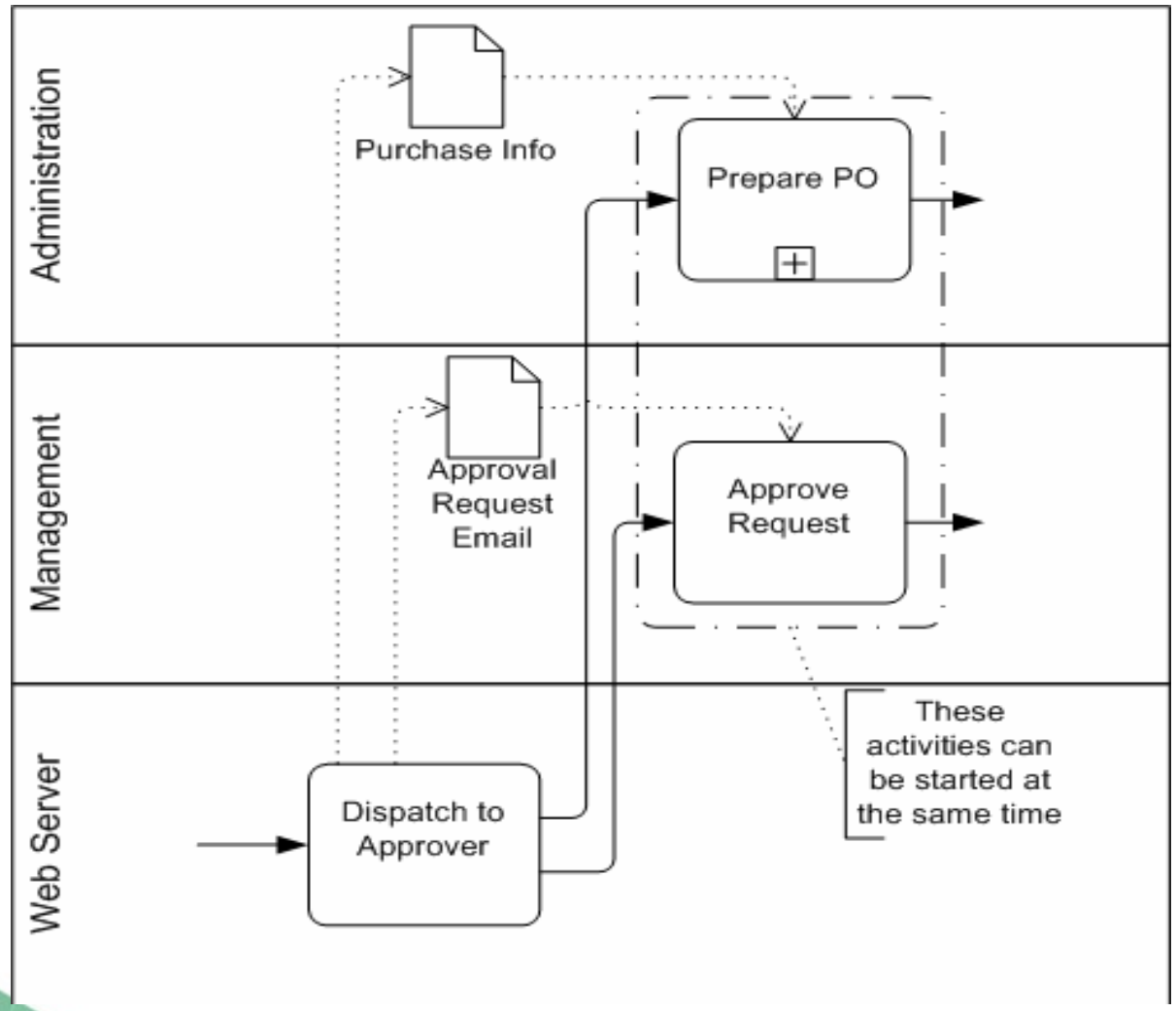
Process Diagram Sample



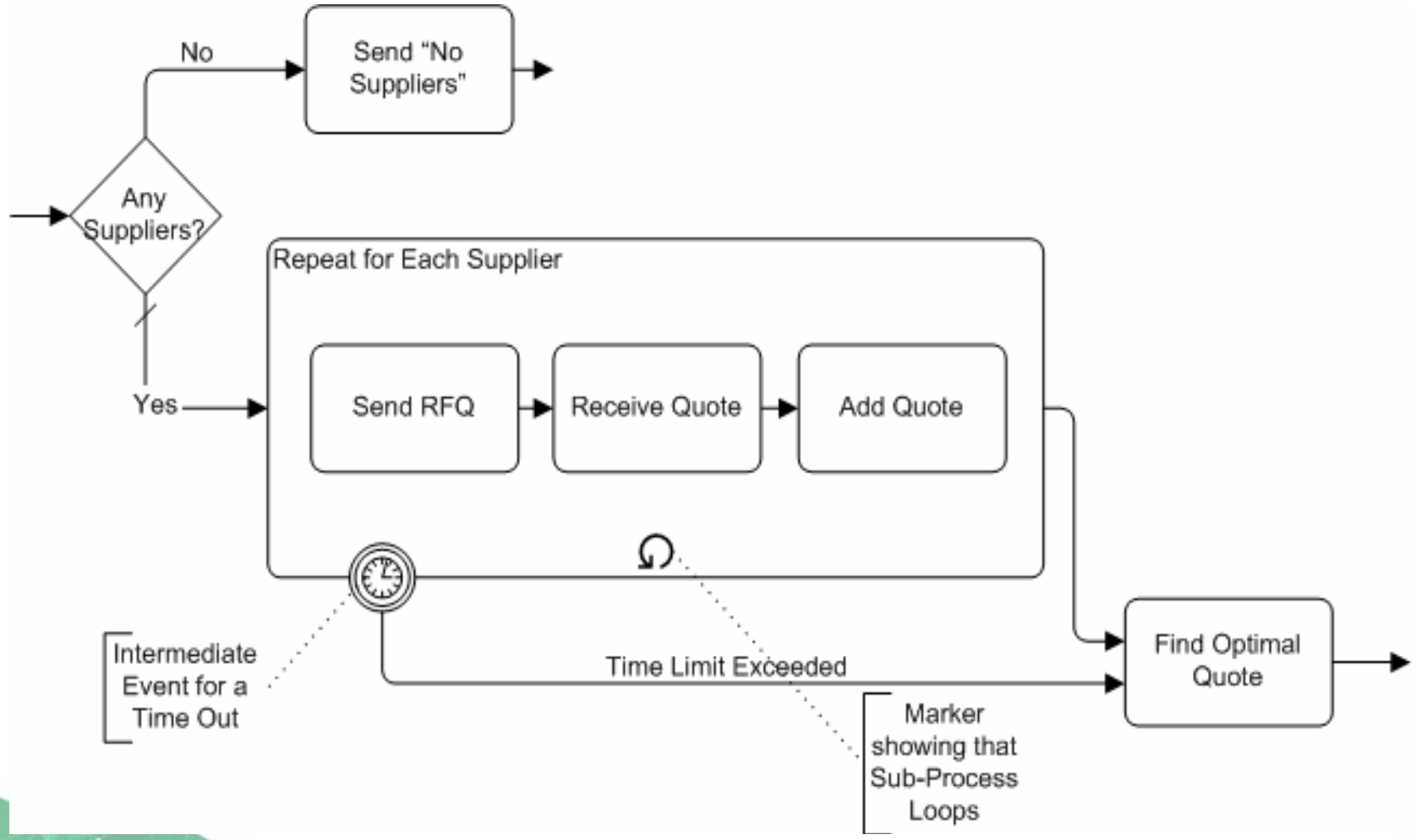
Contoh 2


Patient
Send Doctor Request
Receive Appt.
Send Symptoms
Receive Prescription Pickup
Send Medicine Request
Receive Medicine
Illness Occurs
1) I want to see doctor
5) Go see doctor
6) I feel sick
8) Pickup your medicine and you can leave
9) need my medicine
10) Here is your medicine
Doctor's Office
Receive Doctor Request
Send Appt.
Receive Symptoms
Send Prescription Pickup
Receive Medicine Request
Send Medicine

Contoh 3


Administration
Purchase Info
Prepare PO
Management
Approval Request Email
Approve Request
Web Server
Dispatch to Approver
These activities can be started at the same time

Contoh 4


No
Send "No Suppliers"
Any Suppliers?
Repeat for Each Supplier
Send RFQ
Receive Quote
Add Quote
Yes
Intermediate Event for a Time Out
Time Limit Exceeded
Find Optimal Quote
Marker showing that Sub-Process Loops